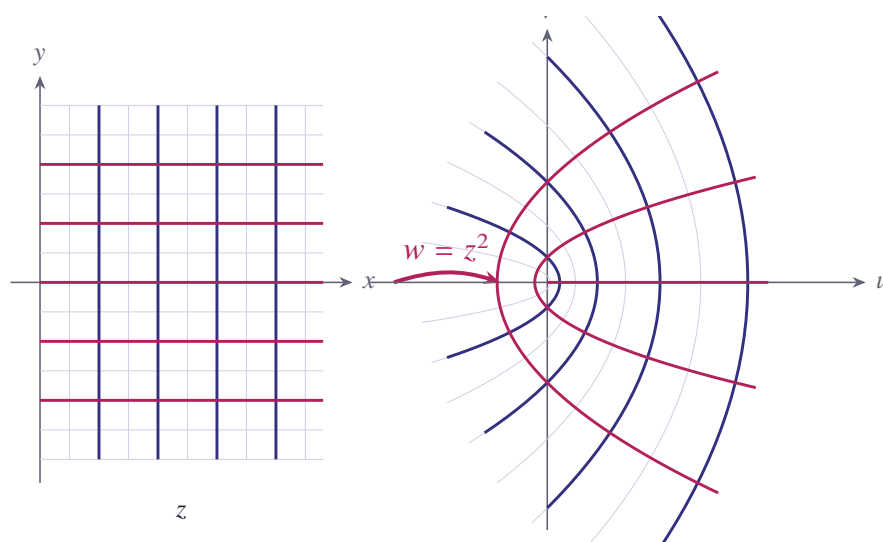


Complex Functions, the Derivative and the Cauchy–Riemann Conditions

توابع مختلط، مشتق و شرایط کشی--ریمان



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9.1 Complex Functions

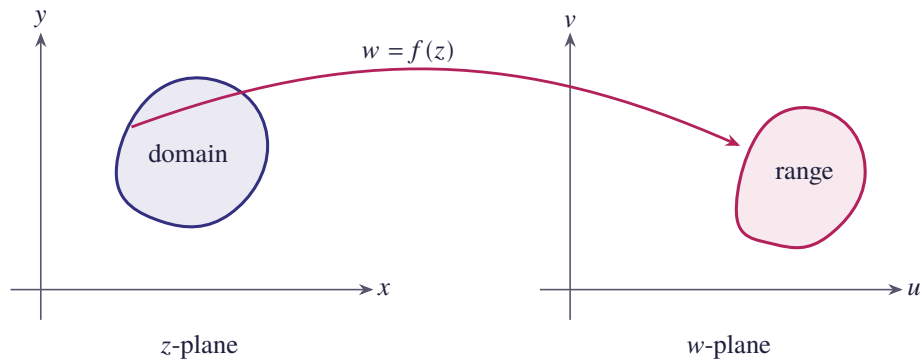


Figure 9.1 A complex function $w = f(z)$ maps points of the domain in the z -plane to points of the range in the w -plane

$$z = x + iy, \quad w = u + iv$$

$$f : \mathbb{C} \rightarrow \mathbb{C}, \quad f : \begin{cases} u = u(x, y) \\ v = v(x, y) \end{cases}$$

Example 9.1

Consider the function $w = z^2$:

$$u + iv = (x + iy)^2 = (x^2 - y^2) + i(2xy) \quad \Rightarrow \quad \begin{cases} u = x^2 - y^2 \\ v = 2xy \end{cases}$$

9.2 Some Important Loci

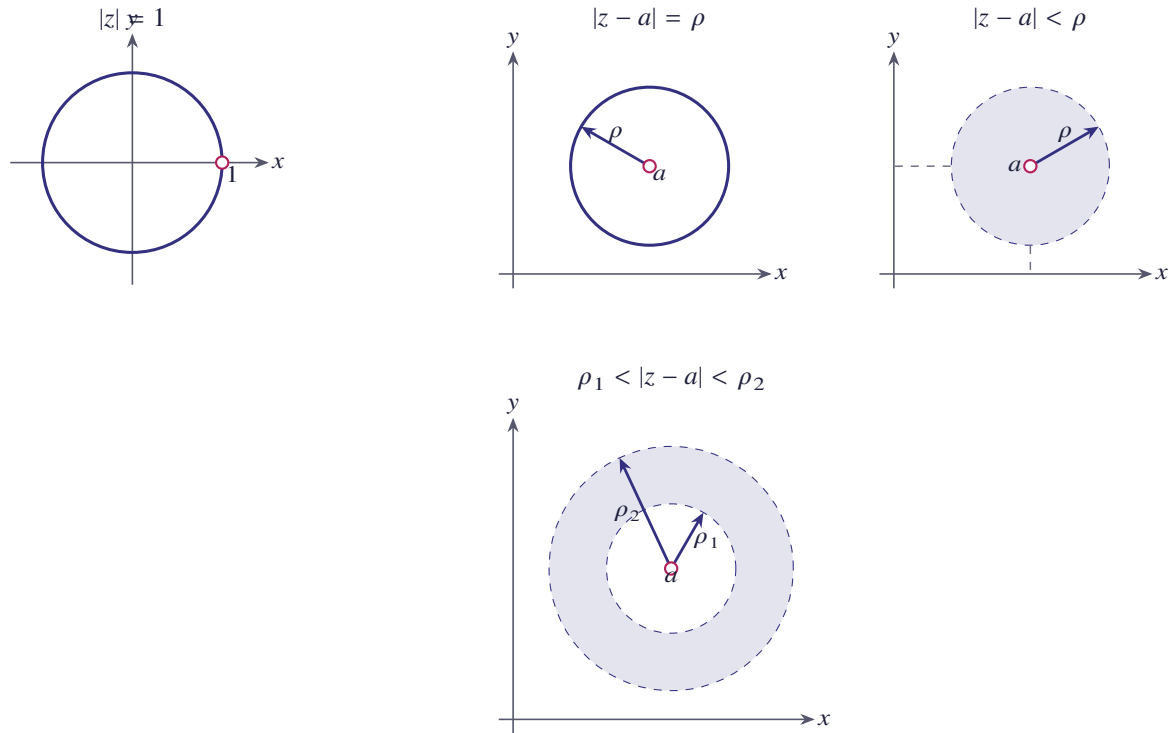


Figure 9.2 Important loci: a circle, the ρ -neighbourhood of a point a , and an annular region

$$|z| = 1, \quad |z - a| = \rho, \quad |z - a| < \rho, \quad \rho_1 < |z - a| < \rho_2$$

The set $|z - a| < \rho$ is called the ρ -neighbourhood of a .

9.3 Limits of Complex Functions

Definition 9.2 ■ Limit

If for every $\varepsilon > 0$ one can find $\delta > 0$ such that $|z - z_0| < \delta$ implies $|f(z) - l| < \varepsilon$, then

$$\lim_{z \rightarrow z_0} f(z) = l$$

Hence for all points inside the circle of centre z_0 and radius δ , $f(z)$ must lie inside the circle of centre l and radius ε . Here ε is given and $\delta(\varepsilon)$ has to be found. Clearly the concept of limit here is broader than that for real functions.

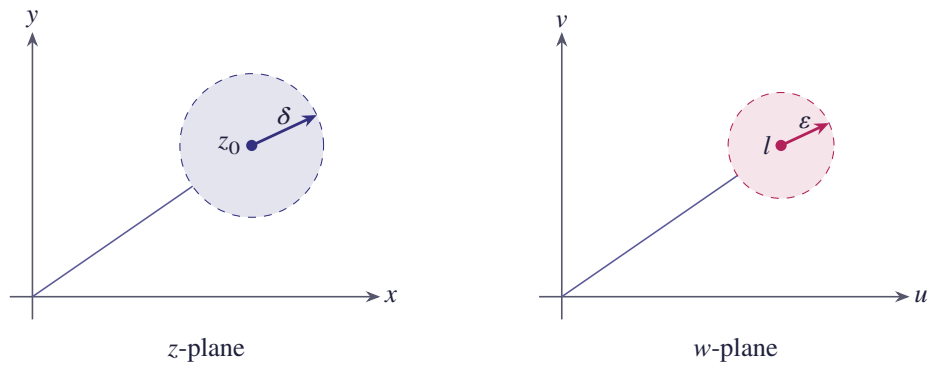


Figure 9.3 The definition of the limit: the δ -neighbourhood of z_0 in the z -plane and the ε -neighbourhood of l in the w -plane

9.4 Continuity of Complex Functions

Definition 9.3 ■ Continuity at a point

The function $w = f(z)$ is continuous at the point $z = z_0$ if the following three conditions hold:

1. $f(z)$ is defined at $z = z_0$.
2. $\lim_{z \rightarrow z_0} f(z)$ exists.
3. $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

Example 9.4

Is the function $f(z) = \sin(1/z)$ continuous at $z = 0$?

Example 9.5

Is the function

$$f(z) = \begin{cases} \sin(1/z) & z \neq 0 \\ 5 + 4i & z = 0 \end{cases}$$

continuous at $z = 0$?

Example 9.6

Is the function

$$f(z) = \begin{cases} 3z^3 + 4 & z \neq 0 \\ 17 & z = 0 \end{cases}$$

continuous at $z = 0$?

Properties of Continuous Complex Functions

Theorem 9.7 ■ Theorems

1. If $f(z)$, $g(z)$ are continuous at z_0 , then $f(z) \pm g(z)$ is continuous at z_0 .
2. If $f(z)$, $g(z)$ are continuous at z_0 , then $f(z) \times g(z)$ is continuous at z_0 .
3. If $f(z)$, $g(z)$ are continuous at z_0 and $g(z_0) \neq 0$, then $\frac{f(z)}{g(z)}$ is continuous at z_0 .

Definition 9.8 ■ Continuity on a domain

The function $w = f(z)$ is continuous on the domain D if f is continuous at every point of D .

Domain: all the points inside a closed curve.

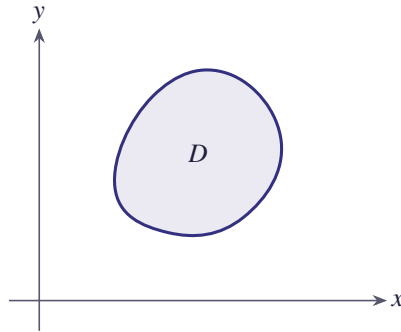


Figure 9.4 A domain D in the complex plane

9.5 The Derivative of Complex Functions

Definition 9.9 ■ Derivative

The function $w = f(z)$ is differentiable at the point $z = z_0$ if the following limit exists:

$$f'(z_0) = \lim_{|\Delta z| \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

Remark

The existence of the derivative of a complex function requires stricter conditions than for real functions: since z_0 can be approached from *all* directions in the z -plane, this limit must exist and must not depend on the chosen path towards z_0 .

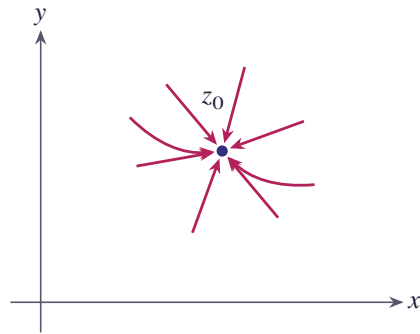


Figure 9.5 Approaching the point z_0 from all directions

9.6 The Cauchy–Riemann Conditions

Theorem 9.10 ■ Necessary condition for differentiability

A necessary condition for the complex function $w = f(z)$ to be differentiable is that the Cauchy–Riemann conditions hold:

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases} \quad w = u + iv = f(x + iy) = f(z)$$

Proof.

$$f'(z) = \lim_{|\Delta z| \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}, \quad z = x + iy, \quad \Delta z = \Delta x + i\Delta y$$

$$\begin{aligned} f(z + \Delta z) - f(z) &= [u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y)] - [u(x, y) + iv(x, y)] \\ &= [u(x + \Delta x, y + \Delta y) - u(x, y)] + i[v(x + \Delta x, y + \Delta y) - v(x, y)] \end{aligned}$$

$$\frac{f(z + \Delta z) - f(z)}{\Delta z} = \frac{u(x + \Delta x, y + \Delta y) - u(x, y)}{\Delta x + i\Delta y} + i \frac{v(x + \Delta x, y + \Delta y) - v(x, y)}{\Delta x + i\Delta y}$$

A necessary condition for the existence of the derivative of $w = f(z)$ is that the value of the limit defining $f'(z)$ does not change if we approach z from any direction (in particular along the horizontal or the vertical direction).

Path 1: the vertical direction. $\Delta z = \Delta x + i\Delta y$ with $\Delta x = 0$, i.e. $\Delta z = i\Delta y$:

$$\begin{aligned} f'(z) &= \lim_{|\Delta z| \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \\ &= \lim_{\Delta y \rightarrow 0} \left[\frac{u(x, y + \Delta y) - u(x, y)}{i\Delta y} + i \frac{v(x, y + \Delta y) - v(x, y)}{i\Delta y} \right] \\ &= \frac{1}{i} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y} \end{aligned}$$

Path 2: the horizontal direction. $\Delta z = \Delta x + i\Delta y$ with $\Delta y = 0$, i.e. $\Delta z = \Delta x$:

$$\begin{aligned} f'(z) &= \lim_{|\Delta z| \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \\ &= \lim_{\Delta x \rightarrow 0} \left[\frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x} \right] \\ &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \end{aligned}$$

Therefore

$$f'(z) = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

and equating real and imaginary parts:

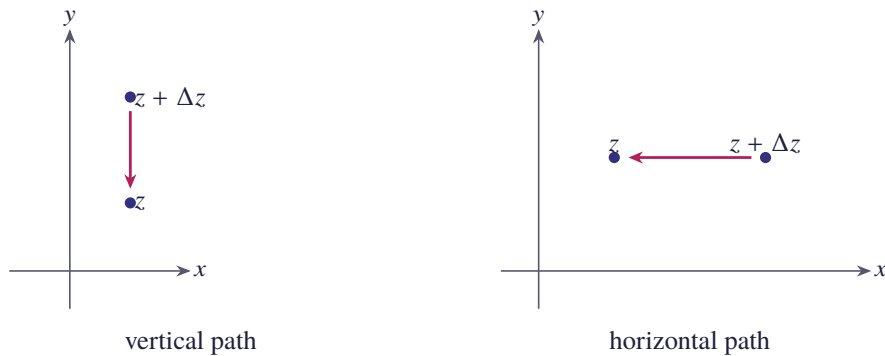


Figure 9.6 Two paths towards z : the vertical direction ($\Delta z = i\Delta y$) and the horizontal direction ($\Delta z = \Delta x$)

Key result 9.11 ■ The Cauchy–Riemann conditions

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Remark ■ Consequence

1. If these conditions do not hold, the derivative does not exist at the point z .
2. The following ways of computing the derivative are suggested:

$$f'(z) = \lim_{|\Delta z| \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} = \frac{\partial v}{\partial y} + i \frac{\partial v}{\partial x}$$

Theorem 9.12 ■ Sufficient condition for differentiability

A sufficient condition for the complex function $w = f(z)$ to be differentiable is that the Cauchy–Riemann conditions hold.

The proof is beyond the scope of this class.

Theorem 9.13 ■ The Cauchy–Riemann theorem

A necessary and sufficient condition for the complex function $w = f(z)$ to be differentiable is that the Cauchy–Riemann conditions hold.

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases} \quad w = u + iv = f(x + iy) = f(z)$$

■ A Consequence of the Cauchy–Riemann Conditions (Analyticity of a Function)

Differentiating the Cauchy–Riemann conditions with respect to x and y :

$$\left. \begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{\partial^2 v}{\partial x \partial y} \\ \frac{\partial^2 u}{\partial y^2} &= -\frac{\partial^2 v}{\partial x \partial y} \end{aligned} \right\} \xrightarrow{\text{add}} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\left. \begin{aligned} \frac{\partial^2 u}{\partial x \partial y} &= \frac{\partial^2 v}{\partial y^2} \\ \frac{\partial^2 u}{\partial x \partial y} &= -\frac{\partial^2 v}{\partial x^2} \end{aligned} \right\} \xrightarrow{\text{subtract}} \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Remark ■ Consequence

The real and imaginary parts of an analytic function satisfy the two-dimensional Laplace equation.

9.7 Differentiability of Complex Functions: Examples

Example 9.14

Consider the function $w = z^2$; is it analytic?

$$w = f(z) = z^2 = (x + iy)^2 = (x^2 - y^2) + i 2xy, \quad u(x, y) = x^2 - y^2, \quad v(x, y) = 2xy$$

$$\frac{\partial u}{\partial x} = 2x = \frac{\partial v}{\partial y} \quad \checkmark \quad \frac{\partial u}{\partial y} = -2y = -\frac{\partial v}{\partial x} \quad \checkmark$$

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = 2x + i 2y = 2(x + iy) = 2z$$

$$\nabla^2 u = 2 - 2 = 0, \quad \nabla^2 v = 0 + 0 = 0$$

Example 9.15

Consider the function $w = |z|^2$; is it analytic?

$$w = f(z) = |z|^2 = (x + iy)(x - iy) = x^2 + y^2, \quad u(x, y) = x^2 + y^2, \quad v(x, y) = 0$$

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial y} = 0, \quad \frac{\partial u}{\partial y} = 2y, \quad -\frac{\partial v}{\partial x} = 0$$

$$\nabla^2 u = 2 + 2 = 4, \quad \nabla^2 v = 0 + 0 = 0$$

Analytic nowhere except at the origin.

Exercise 9.16

Determine whether each of the following functions is analytic. When it is, find its derivative.

1. $f(z) = \bar{z}$
2. $f(z) = \operatorname{Re}\{z\}$
3. $f(z) = z^3 + 2z^2 + 3i$

9.8 Analytic Complex Functions

Example 9.17

$u = x^2 - y^2$ is the real part of an analytic function $f(z)$. Find $f(z)$.

Since $f(z)$ is analytic, its real and imaginary parts satisfy the Cauchy–Riemann conditions, and consequently they must also satisfy the two-dimensional Laplace equation.

- 1) First check whether the real part satisfies the two-dimensional Laplace equation:

$$\nabla^2 u = \frac{\partial^2}{\partial x^2}(x^2 - y^2) + \frac{\partial^2}{\partial y^2}(x^2 - y^2) = 2 - 2 = 0 \quad \checkmark$$

- 2) We use the Cauchy–Riemann relations:

$$\begin{aligned} \frac{\partial u}{\partial x} = 2x = \frac{\partial v}{\partial y} &\implies v(x, y) = 2xy + \varphi(x) \\ -\frac{\partial v}{\partial x} = -2y - \varphi'(x) = \frac{\partial u}{\partial y} = -2y &\implies \varphi'(x) = 0 \implies \varphi(x) = c, \quad c \in \mathbb{R} \\ f(z) = (x^2 - y^2) + i(2xy + c) &= z^2 + C \end{aligned}$$

Example 9.18

Let $u = \cos x \cosh y$. (a) Show that this function is harmonic in the plane ($\nabla^2 u = 0$). (b) Find $v(x, y)$ such that $u + iv$ is analytic.

- (a) Check whether the real part satisfies the two-dimensional Laplace equation:

$$\nabla^2 u = \frac{\partial^2}{\partial x^2}(\cos x \cosh y) + \frac{\partial^2}{\partial y^2}(\cos x \cosh y) = -\cos x \cosh y + \cos x \cosh y = 0 \quad \checkmark$$

- (b) We use the Cauchy–Riemann relations:

$$\begin{aligned} \frac{\partial u}{\partial x} = -\sin x \cosh y = \frac{\partial v}{\partial y} &\implies v(x, y) = -\sin x \sinh y + \varphi(x) \\ -\frac{\partial v}{\partial x} = \cos x \sinh y - \varphi'(x) = \frac{\partial u}{\partial y} = \cos x \sinh y &\implies \varphi'(x) = 0 \\ \varphi(x) = c, \quad c \in \mathbb{R} &\implies v(x, y) = -\sin x \sinh y + c \end{aligned}$$

9.9 The Cauchy–Riemann Conditions in Polar Coordinates

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \quad \begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \tan^{-1} \frac{y}{x} \end{cases} \quad r_x = \frac{x}{r}, \quad \theta_x = \frac{-y}{r^2}, \quad r_y = \frac{y}{r}, \quad \theta_y = \frac{x}{r^2}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x} = \frac{x}{r} \frac{\partial u}{\partial r} - \frac{y}{r^2} \frac{\partial u}{\partial \theta} = \cos \theta \frac{\partial u}{\partial r} - \frac{1}{r} \sin \theta \frac{\partial u}{\partial \theta}$$

$$\frac{\partial v}{\partial y} = \frac{\partial v}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial v}{\partial \theta} \frac{\partial \theta}{\partial y} = \frac{y}{r} \frac{\partial v}{\partial r} + \frac{x}{r^2} \frac{\partial v}{\partial \theta} = \sin \theta \frac{\partial v}{\partial r} + \frac{1}{r} \cos \theta \frac{\partial v}{\partial \theta}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \implies \cos \theta \frac{\partial u}{\partial r} - \frac{1}{r} \sin \theta \frac{\partial u}{\partial \theta} = \sin \theta \frac{\partial v}{\partial r} + \frac{1}{r} \cos \theta \frac{\partial v}{\partial \theta}$$

Similarly

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \Rightarrow \sin \theta \frac{\partial u}{\partial r} + \frac{1}{r} \cos \theta \frac{\partial u}{\partial \theta} = -\cos \theta \frac{\partial v}{\partial r} + \frac{1}{r} \sin \theta \frac{\partial v}{\partial \theta}$$

Key result 9.19 ■ The Cauchy–Riemann conditions in polar form

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad -\frac{1}{r} \frac{\partial u}{\partial \theta} = \frac{\partial v}{\partial r}$$

Exercise 9.20

Using the Cauchy–Riemann conditions in polar form, prove that u and v satisfy Laplace’s equation in polar form, i.e.

$$\nabla^2 u = u_{rr} + \frac{1}{r^2} u_{\theta\theta} + \frac{1}{r} u_r = 0, \quad \nabla^2 v = v_{rr} + \frac{1}{r^2} v_{\theta\theta} + \frac{1}{r} v_r = 0$$